

input sequence - i am tired, i want to

Solution: i, am, tired, want

Embeddings: i = 0.2 am = 0.3 tired = 0.8 want = 0.6

initial hidden : $h_0 = 0$

weights

$W_z=0.5$ $U_z=0.1$ (U_z , U_r , U_h are **weights for the hidden state (memory part)** in a GRU)

$W_r=0.4$ $U_r=0.2$

$W_h=0.6$ $U_h=0.3$

GRU equations:

update gate $z = \text{sigmoid}(W_z \cdot x + U_z \cdot h_{\text{prev}})$

reset gate $r = \text{sigmoid}(W_r \cdot x + U_r \cdot h_{\text{prev}})$

candidate hidden state $\tilde{h} = \tanh(W_h \cdot x + U_h \cdot (r \cdot h_{\text{prev}}))$

final hidden state $h = (1 - z) \cdot h_{\text{prev}} + z \cdot \tilde{h}$

STEP 1: word “i” ($x=0.2$, $h_{\text{prev}}=0$)

$$z = \text{sigmoid}(0.5 \times 0.2 + 0.1 \times 0) = \text{sigmoid}(0.1) = 1 / (1 + e^{-0.1}) \approx 1 / (1 + 0.904) \approx 0.52$$

$$r = \text{sigmoid}(0.4 \times 0.2 + 0.2 \times 0) = \text{sigmoid}(0.08) \approx 0.52$$

$$\tilde{h} = \tanh(0.6 \times 0.2 + 0.3 \times (0.52 \times 0)) = \tanh(0.12) \approx 0.12$$

$$h_1 = (1 - 0.52) \times 0 + 0.52 \times 0.12 = 0.062$$

STEP 2: “am” ($x=0.3$, $h_{\text{prev}}=0.062$)

$$z = \text{sigmoid}(0.5 \times 0.3 + 0.1 \times 0.062) = \text{sigmoid}(0.1562) \approx 0.54$$

$$r = \text{sigmoid}(0.4 \times 0.3 + 0.2 \times 0.062) = \text{sigmoid}(0.1324) \approx 0.53$$

$$\tilde{h} = \tanh(0.6 \times 0.3 + 0.3 \times (0.53 \times 0.062)) = \tanh(0.1898) \approx 0.19$$

$$h_2 = (1 - 0.54) \times 0.062 + 0.54 \times 0.19 = 0.0285 + 0.1026 = 0.131$$

STEP 3: “tired” ($x=0.8$, $h_{\text{prev}}=0.131$)

$$z = \text{sigmoid}(0.5 \times 0.8 + 0.1 \times 0.131) = \text{sigmoid}(0.4131) \approx 0.60$$

$$r = \text{sigmoid}(0.4 \times 0.8 + 0.2 \times 0.131) = \text{sigmoid}(0.3462) \approx 0.5$$

$$\tilde{h} = \tanh(0.6 \times 0.8 + 0.3 \times (0.5 \times 0.131)) = \tanh(0.5032) \approx 0.46$$

$$h_3 = (1 - 0.60) \times 0.131 + 0.60 \times 0.46 = 0.0524 + 0.276 = 0.328$$

STEP 4: “want” ($x=0.6$, $h_{\text{prev}}=0.328$)

$$z = \text{sigmoid}(0.5 \times 0.6 + 0.1 \times 0.328) = \text{sigmoid}(0.3328) \approx 0.58$$

$$r = \text{sigmoid}(0.4 \times 0.6 + 0.2 \times 0.328) = \text{sigmoid}(0.3056) \approx 0.58$$

$$\tilde{h} = \tanh(0.6 \times 0.6 + 0.3 \times (0.58 \times 0.328)) = \tanh(0.417) \approx 0.39$$

$$h_4 = (1 - 0.58) \times 0.328 + 0.58 \times 0.39 = 0.1377 + 0.226 = 0.364$$

FINAL HIDDEN STATE

$$h = 0.364$$

In GRU step, it **looks like nothing is forgotten**, but actually forgetting is happening **implicitly**

GRU does NOT have a separate “forget gate” like LSTM, it uses **update gate z** to both **remember + forget**

$$\text{Final equation } h = (1 - z) \cdot h_{\text{prev}} + z \cdot \tilde{h}$$

$$z = 0.58, 1 - z = 0.42$$

$$\begin{aligned} \text{So } h &= 0.42 \times h_{\text{prev}} + 0.58 \times \text{new_info} \\ &= 0.42 \times 0.328 + 0.58 \times 0.39 \end{aligned}$$

what does this mean? $0.42 \rightarrow$ only 42% of old memory kept, $58\% \rightarrow$ replaced with new information

so actually 58% of old info is forgotten

so where is forgetting?

it is here: $(1 - z) \cdot h_{\text{prev}}$

if z is high $\rightarrow$ more forgetting

if z is low $\rightarrow$ less forgetting

quick intuition $z \approx 1 \rightarrow$ forget almost everything, use new info

$z \approx 0 \rightarrow$ keep old memory, ignore new info

what about reset gate r ? that works earlier: $\tilde{h} = \tanh(W_h \cdot x + U_h \cdot (r \cdot h_{\text{prev}}))$

in our case $r = 0.58$, so only 58% of h_{prev} is used to create new candidate

rest is ignored $\rightarrow$ partial forgetting happens here too

so GRU has TWO forgetting places - r gate $\rightarrow$ controls how much past is used to create new info

z gate $\rightarrow$ controls how much old memory is replaced

compare with LSTM

LSTM: explicit forget gate f

GRU: no f , but z does the job

in our example

old memory = 0.328, final = 0.364

it looks similar $\rightarrow$ because z not very high

but still old memory contribution reduced from 0.328 $\rightarrow$ 0.137, so more than half influence is replaced

PREDICTION STEP: assume output words sleep, eat, run

Weights sleep = 1.5 eat = 1.0 run = 0.5

Scores : sleep = $1.5 \times 0.364 = 0.546$, eat = $1.0 \times 0.364 = 0.364$, run = $0.5 \times 0.364 = 0.182$

SOFTMAX

$e^{0.546} \approx 1.73$

$e^{0.364} \approx 1.44$

$e^{0.182} \approx 1.20$

sum = $1.73 + 1.44 + 1.20 = 4.37$

probabilities

sleep = $1.73 / 4.37 = 0.40$

eat = $1.44 / 4.37 = 0.33$

run = $1.20 / 4.37 = 0.27$

FINAL ANSWER

highest probability = sleep

Prediction i am tired i want to sleep

LSTM solution

i=0.2, am=0.3, tired=0.8, want=0.6, **h0=0, C0=0**, Wf=0.5, Uf=0.1, Wi=0.6, Ui=0.2, Wo=0.4, Uo=0.1, Wc=0.7, Uc=0.3

EQUATIONS

f = sigmoid(Wf·x + Uf·h_prev)
i = sigmoid(Wi·x + Ui·h_prev)
o = sigmoid(Wo·x + Uo·h_prev)
C~ = tanh(Wc·x + Uc·h_prev)

C = f·C_prev + i·C~
h = o·tanh(C)

step1(i): f=sigmoid(0.5×0.2+0.1×0)=sigmoid(0.1)=0.52,
i=sigmoid(0.6×0.2+0.2×0)=sigmoid(0.12)=0.53,
o=sigmoid(0.4×0.2+0.1×0)=sigmoid(0.08)=0.52,
C~=tanh(0.7×0.2+0.3×0)=tanh(0.14)=0.14, C1=0.52×0+0.53×0.14=0.074,
h1=0.52×tanh(0.074)=0.038

step2(am): f=sigmoid(0.5×0.3+0.1×0.038)=sigmoid(0.1538)=0.54,
i=sigmoid(0.6×0.3+0.2×0.038)=sigmoid(0.1876)=0.55,
o=sigmoid(0.4×0.3+0.1×0.038)=sigmoid(0.1238)=0.53,
C~=tanh(0.7×0.3+0.3×0.038)=tanh(0.2214)=0.22, C2=0.54×0.074+0.55×0.22=0.161,
h2=0.53×tanh(0.161)=0.084

step3(tired): f=sigmoid(0.5×0.8+0.1×0.084)=sigmoid(0.4084)=0.60,
i=sigmoid(0.6×0.8+0.2×0.084)=sigmoid(0.4968)=0.62,
o=sigmoid(0.4×0.8+0.1×0.084)=sigmoid(0.3284)=0.58,
C~=tanh(0.7×0.8+0.3×0.084)=tanh(0.5852)=0.53, C3=0.60×0.161+0.62×0.53=0.425,
h3=0.58×tanh(0.425)=0.232

step4(want): f=sigmoid(0.5×0.6+0.1×0.232)=sigmoid(0.3232)=0.58,
i=sigmoid(0.6×0.6+0.2×0.232)=sigmoid(0.4064)=0.60,
o=sigmoid(0.4×0.6+0.1×0.232)=sigmoid(0.2632)=0.56,
C~=tanh(0.7×0.6+0.3×0.232)=tanh(0.4896)=0.45, C4=0.58×0.425+0.60×0.45=0.516,
h4=0.56×tanh(0.516)=0.263

final: C=0.516, h=0.263

prediction: sleep= $1.5 \times 0.263 = 0.395$, eat= $1.0 \times 0.263 = 0.263$, run= $0.5 \times 0.263 = 0.132$,
softmax $\rightarrow$ sleep highest so output = i am tired i want to sleep

stock prices $x_1=100$, $x_2=102$, $x_3=101$, $x_4=105$, normalize ($/200$) $\rightarrow x_1=0.5$, $x_2=0.51$,
 $x_3=0.505$, $x_4=0.525$, initial $h_0=0.3$, $W_z=0.5$ $U_z=0.1$ $W_r=0.4$ $U_r=0.2$ $W_h=0.6$ $U_h=0.3$

For normalization taking highest value 200 so that values are between 0 and 1

step1($x_1=0.5$): $z = \text{sigmoid}(0.5 \times 0.5 + 0.1 \times 0.3) = \text{sigmoid}(0.25 + 0.03) = \text{sigmoid}(0.28) = 0.57$,
 $r = \text{sigmoid}(0.4 \times 0.5 + 0.2 \times 0.3) = \text{sigmoid}(0.2 + 0.06) = \text{sigmoid}(0.26) = 0.56$,
 $h = \tanh(0.6 \times 0.5 + 0.3 \times (0.56 \times 0.3)) = \tanh(0.3 + 0.0504) = \tanh(0.3504) = 0.34$,
 $h_1 = (1 - 0.57) \times 0.3 + 0.57 \times 0.34 = 0.43 \times 0.3 + 0.1938 = 0.129 + 0.1938 = 0.323$

step2($x_2=0.51$):
 $z = \text{sigmoid}(0.5 \times 0.51 + 0.1 \times 0.323) = \text{sigmoid}(0.255 + 0.0323) = \text{sigmoid}(0.2873) = 0.57$,
 $r = \text{sigmoid}(0.4 \times 0.51 + 0.2 \times 0.323) = \text{sigmoid}(0.204 + 0.0646) = \text{sigmoid}(0.2686) = 0.57$,
 $h = \tanh(0.6 \times 0.51 + 0.3 \times (0.57 \times 0.323)) = \tanh(0.306 + 0.0552) = \tanh(0.3612) = 0.35$,
 $h_2 = (1 - 0.57) \times 0.323 + 0.57 \times 0.35 = 0.43 \times 0.323 + 0.1995 = 0.139 + 0.1995 = 0.339$

step3($x_3=0.505$):
 $z = \text{sigmoid}(0.5 \times 0.505 + 0.1 \times 0.339) = \text{sigmoid}(0.2525 + 0.0339) = \text{sigmoid}(0.2864) = 0.57$,
 $r = \text{sigmoid}(0.4 \times 0.505 + 0.2 \times 0.339) = \text{sigmoid}(0.202 + 0.0678) = \text{sigmoid}(0.2698) = 0.57$,
 $h = \tanh(0.6 \times 0.505 + 0.3 \times (0.57 \times 0.339)) = \tanh(0.303 + 0.0580) = \tanh(0.361) = 0.35$,
 $h_3 = (1 - 0.57) \times 0.339 + 0.57 \times 0.35 = 0.43 \times 0.339 + 0.1995 = 0.146 + 0.1995 = 0.345$

step4($x_4=0.525$):
 $z = \text{sigmoid}(0.5 \times 0.525 + 0.1 \times 0.345) = \text{sigmoid}(0.2625 + 0.0345) = \text{sigmoid}(0.297) = 0.57$,
 $r = \text{sigmoid}(0.4 \times 0.525 + 0.2 \times 0.345) = \text{sigmoid}(0.21 + 0.069) = \text{sigmoid}(0.279) = 0.57$,
 $h = \tanh(0.6 \times 0.525 + 0.3 \times (0.57 \times 0.345)) = \tanh(0.315 + 0.0590) = \tanh(0.374) = 0.36$,
 $h_4 = (1 - 0.57) \times 0.345 + 0.57 \times 0.36 = 0.43 \times 0.345 + 0.2052 = 0.148 + 0.2052 = 0.353$

prediction layer $W_y=1.3$ $b=0.1$ (Assumed)

$y = 1.3 \times 0.353 + 0.1 = 0.459 + 0.1 = 0.559$

final predicted normalized price $\approx 0.559 \rightarrow \text{rescale} \times 200 \approx 111.8 \approx 112$ (next stock price)

W_y and b are found by minimizing error using gradient descent, not assumed

given $h=0.353$, true price=110 $\rightarrow y_true=110/200=0.55$, initial $W_y=1$, $b=0$,
 step1 prediction: $y_pred=W_y \times h + b = 1 \times 0.353 + 0 = 0.353$,
 step2 loss: $loss=(y_pred-y_true)^2=(0.353-0.55)^2=(-0.197)^2=0.039$,
 step3 gradients: $dL/dy=2(y_pred-y_true)=2(0.353-0.55)=-0.394$,
 $dL/dW_y=dL/dy \times h=-0.394 \times 0.353=-0.139$, $dL/db=dL/dy=-0.394$, step4 update ($lr=0.1$):
 $W_{y_new}=1-0.1 \times (-0.139)=1.014$, $b_{new}=0-0.1 \times (-0.394)=0.0394$,
 step5 new prediction: $y_pred=1.014 \times 0.353 + 0.0394=0.358+0.0394=0.397$

Multivariate GRU

dataset (3 timesteps): $t_1(\text{price}=100, \text{volume}=1500)$, $t_2(\text{price}=102, \text{volume}=1800)$,
 $t_3(\text{price}=101, \text{volume}=1600)$, normalize ($/200, /5000$) $\rightarrow t_1=[0.5, 0.3]$, $t_2=[0.51, 0.36]$,
 $t_3=[0.505, 0.32]$, initial $h_0=[0.2, 0.1]$, weights $W_z=[[0.5, 0.2], [0.1, 0.4]]$,
 $U_z=[[0.1, 0.3], [0.2, 0.1]]$, $W_r=[[0.4, 0.1], [0.3, 0.2]]$, $U_r=[[0.2, 0.1], [0.1, 0.3]]$,
 $W_h=[[0.6, 0.2], [0.1, 0.5]]$, $U_h=[[0.3, 0.2], [0.2, 0.4]]$

$$X = [[0.5, 0.3], [0.51, 0.36], [0.505, 0.32]]^T, h_0 = [0.2, 0.1]^T$$

$$W_z = [[0.5, 0.2], [0.1, 0.4]], U_z = [[0.1, 0.3], [0.2, 0.1]]$$

$$W_r = [[0.4, 0.1], [0.3, 0.2]], U_r = [[0.2, 0.1], [0.1, 0.3]]$$

$$W_h = [[0.6, 0.2], [0.1, 0.5]], U_h = [[0.3, 0.2], [0.2, 0.4]]$$

$$t=1: x_1=[0.5, 0.3]^T$$

$$z_1 = \text{sigmoid}(W_z \cdot x_1 + U_z \cdot h_0) = \text{sigmoid}([0.31, 0.17]^T + [0.05, 0.05]^T) =$$

$$\text{sigmoid}([0.36, 0.22]^T) \approx [0.59, 0.55]^T$$

$$r_1 = \text{sigmoid}(W_r \cdot x_1 + U_r \cdot h_0) = \text{sigmoid}([0.23, 0.21]^T + [0.05, 0.05]^T) =$$

$$\text{sigmoid}([0.28, 0.26]^T) \approx [0.57, 0.56]^T$$

$$h_{\sim 1} = \tanh(W_h \cdot x_1 + U_h \cdot (r_1 \odot h_0)) = \tanh([0.36, 0.20]^T + [0.045, 0.045]^T) =$$

$$\tanh([0.405, 0.245]^T) \approx [0.38, 0.24]^T$$

$$h_1 = (1 - z_1) \odot h_0 + z_1 \odot h_{\sim 1} \approx [0.306, 0.177]^T$$

$$t=2: x_2=[0.51, 0.36]^T$$

$$z_2 = \text{sigmoid}(W_z \cdot x_2 + U_z \cdot h_1) \approx \text{sigmoid}([0.327, 0.195]^T + [0.084, 0.079]^T) \approx [0.60, 0.57]^T$$

$$r_2 = \text{sigmoid}(W_r \cdot x_2 + U_r \cdot h_1) \approx \text{sigmoid}([0.24, 0.225]^T + [0.079, 0.084]^T) \approx [0.58, 0.58]^T$$

$$h_{\sim 2} = \tanh(W_h \cdot x_2 + U_h \cdot (r_2 \odot h_1)) \approx \tanh([0.378, 0.231]^T + [0.074, 0.077]^T) \approx [0.42, 0.30]^T$$

$$h_2 = (1 - z_2) \odot h_1 + z_2 \odot h_{\sim 2} \approx [0.374, 0.247]^T$$

$t=3: x_3=[0.505,0.32]^T$

$z_3 = \text{sigmoid}(Wz \cdot x_3 + Uz \cdot h_2) \approx [0.61,0.57]^T$

$r_3 = \text{sigmoid}(Wr \cdot x_3 + Ur \cdot h_2) \approx [0.58,0.58]^T$

$h_{\sim 3} = \tanh(Wh \cdot x_3 + Uh \cdot (r_3 \odot h_2)) \approx [0.43,0.30]^T$

$h_3 = (1-z_3) \odot h_2 + z_3 \odot h_{\sim 3} \approx [0.408,0.277]^T$

prediction: $Wy=[1.2,0.8]$, $b=0.1$ (Assumed)

$y = Wy \cdot h_3 + b = [1.2,0.8] \cdot [0.408,0.277] + 0.1 = 0.490 + 0.222 + 0.1 = 0.812$

final normalized prediction = 0.812 $\rightarrow$ actual ≈ 162

Examples of **embeddings in NLP (transformers)** in simple numeric style, no lines

word embeddings: “i am tired” $\rightarrow$ tokens=[i,am,tired], embeddings $\rightarrow$

i=[0.12,-0.45,0.88], am=[0.30,0.11,-0.22], tired=[0.91,0.77,-0.12]

positional embeddings (for order): positions=[1,2,3], pos vectors $\rightarrow$ p1=[0.01,0.02,0.03],

p2=[0.02,0.04,0.06], p3=[0.03,0.06,0.09]

final input to transformer = word + position $\rightarrow$

i=[0.12+0.01,-0.45+0.02,0.88+0.03]=[0.13,-0.43,0.91], am=[0.32,0.15,-0.16],

tired=[0.94,0.83,-0.03]

contextual embeddings (after transformer layer): i=[0.20,-0.30,0.70],

am=[0.40,0.10,-0.10], tired=[1.10,0.95,0.20]

subword embeddings (BPE example): “playing” $\rightarrow$ [play, ##ing], play=[0.5,0.2],

##ing=[0.3,0.1] $\rightarrow$ combined representation used

sentence embedding (CLS token): [CLS]=[0.75,0.60,0.55]

Different types summary: word embedding, positional embedding, segment embedding (sentence A/B), contextual embedding (output of transformer), subword embedding

one line idea: transformer input = token embedding + positional embedding $\rightarrow$ output = contextual embedding

exposure bias is the mismatch where decoder is trained using ground truth inputs (teacher forcing) but during inference uses its own predictions, causing error accumulation